5. REPLACEMENT GATE FABRICATION RESULT

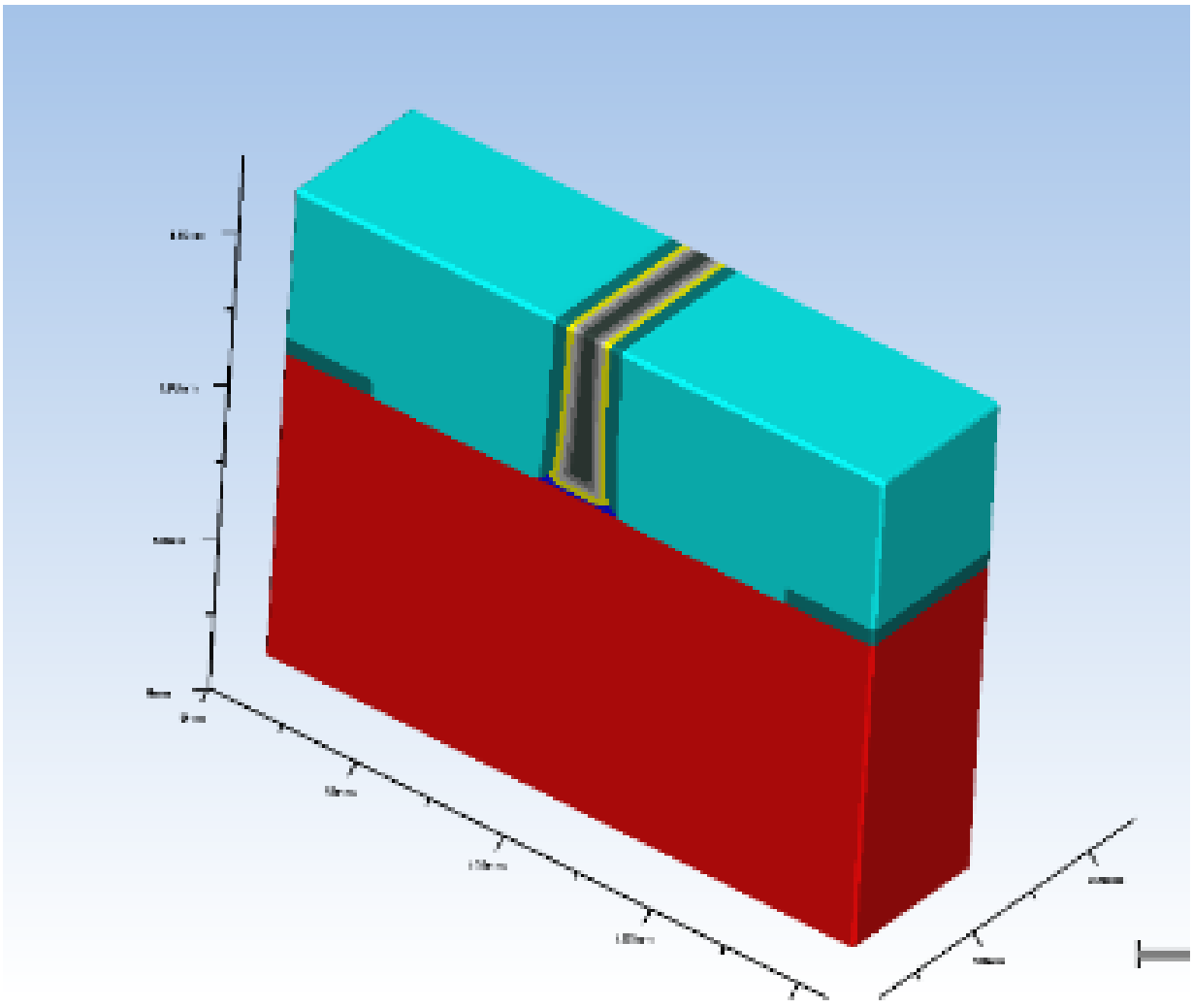


Figure 1: Final 3D Structure of Replacement Metal Gate (RMG) Process

7. ALUMINIUM CONCENTRATION FIELD

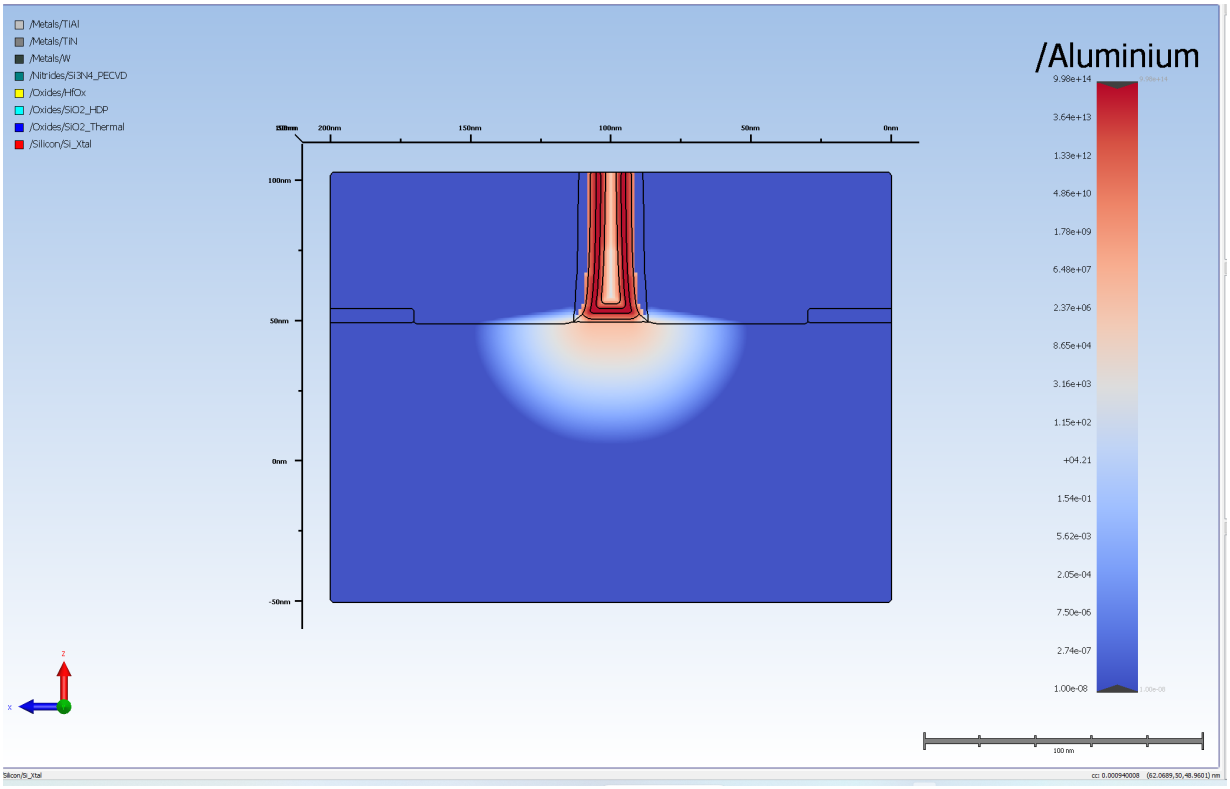


Figure 2: Aluminium Concentration Field Color Map After Dummy Annealing

9. GATE CD VIRTUAL METROLOGY RESPONSE

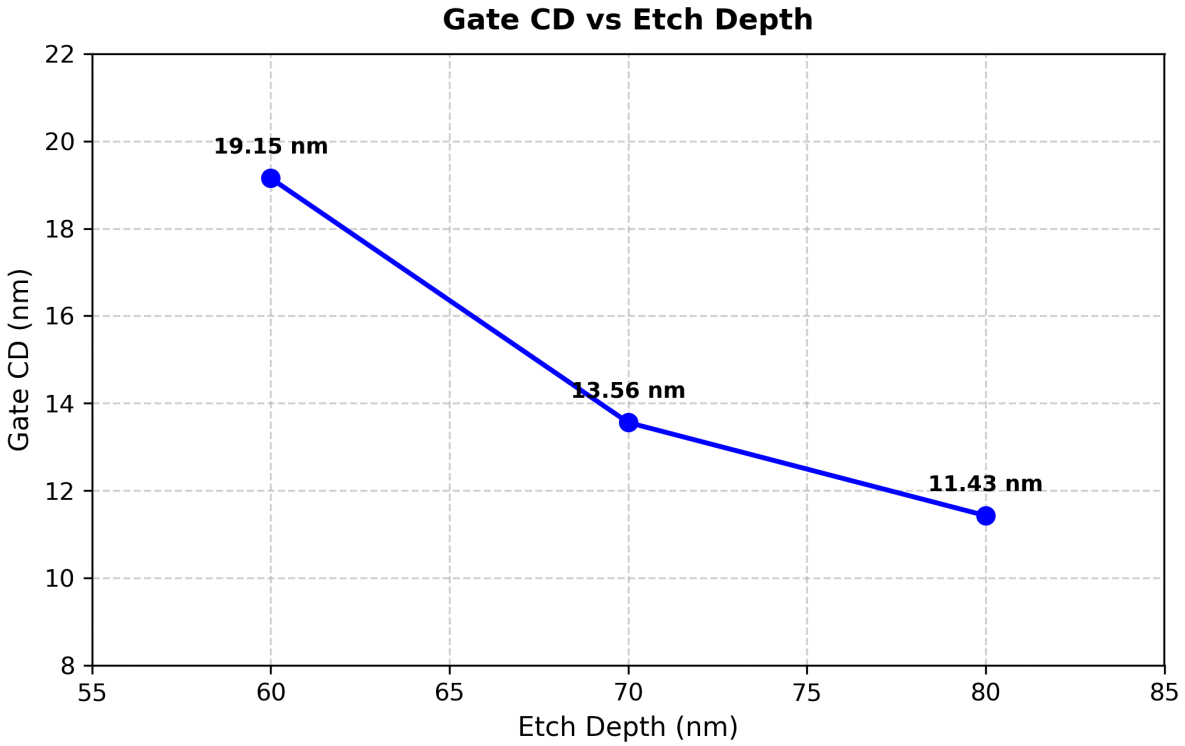


Figure 3: Gate Critical Dimension (CD) vs. Gate Etch Depth

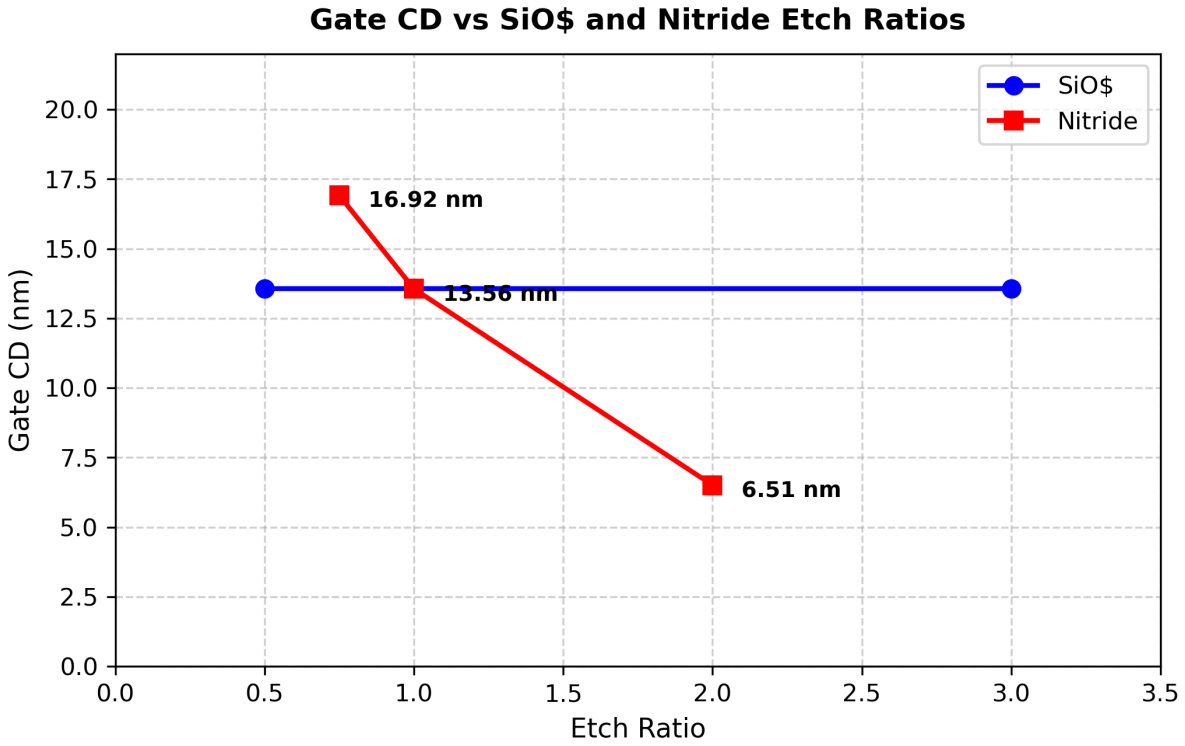


Figure 4: Gate Critical Dimension (CD) vs. SiO2 and Nitride Etch Ratios

11. ALUMINIUM PROFILE: 5 NM LITHO BIAS

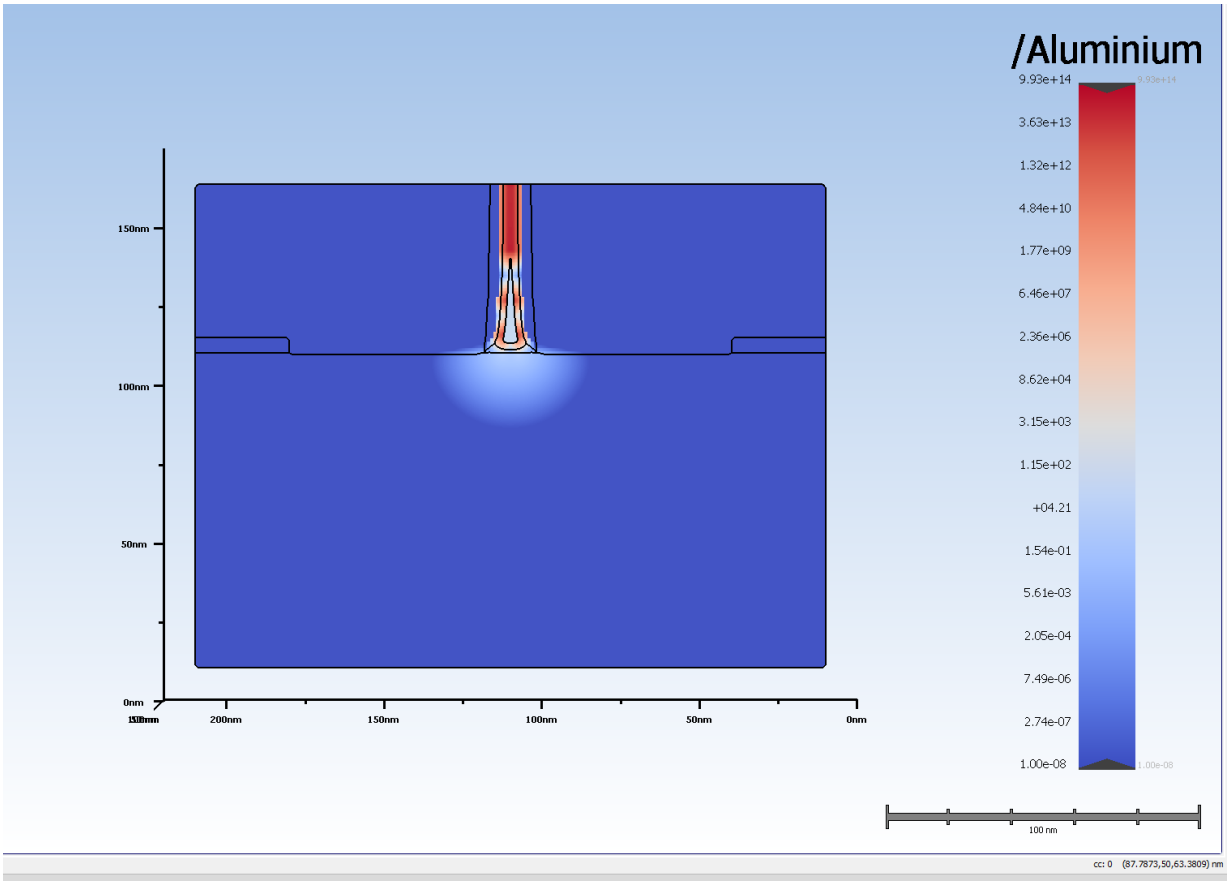


Figure 5: Aluminium Concentration Field After 5 nm Lithography Bias

12. GATE CAVITY VOID DEFECT ANALYSIS

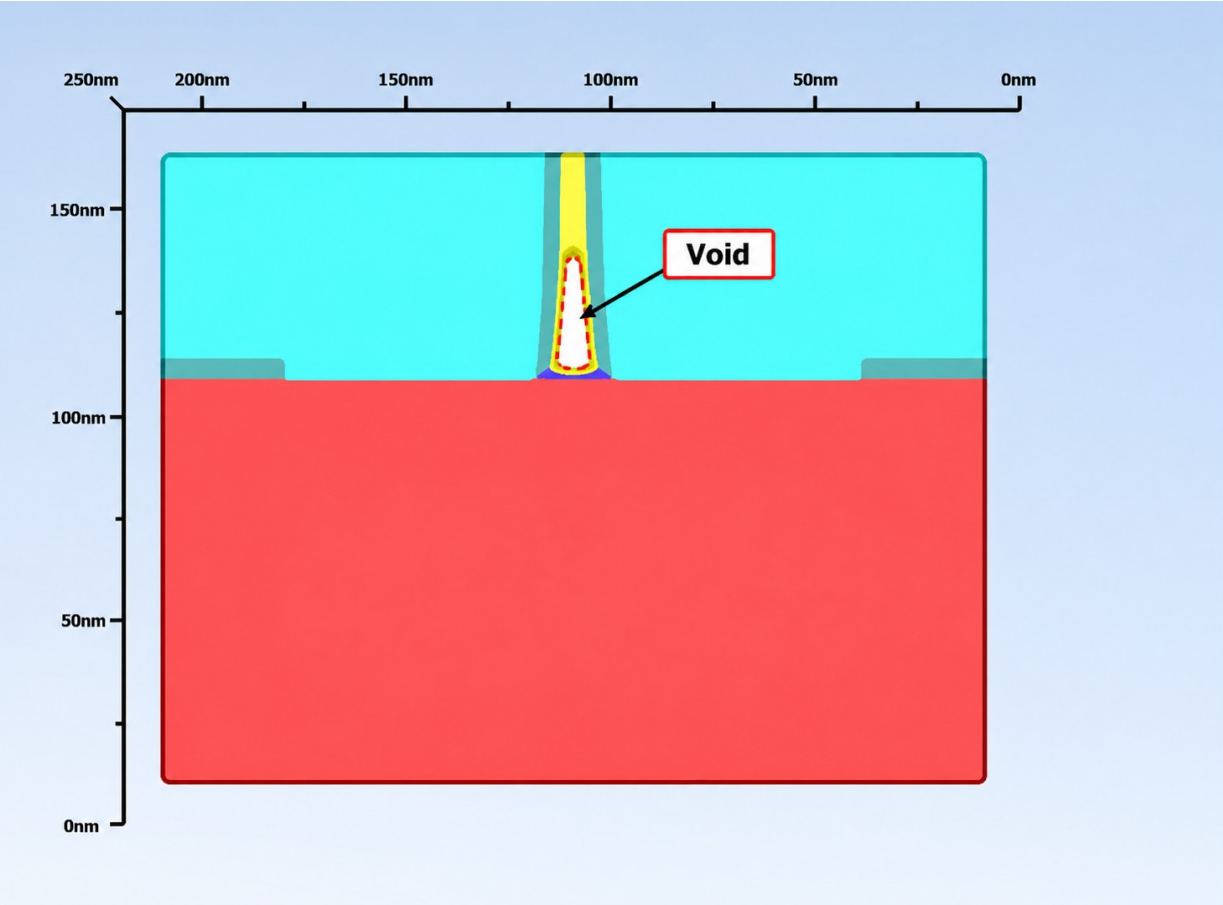


Figure 6: Void Defect Formation Inside Gate Cavity (SiO2 Etch Ratio = 2)

13. ALUMINIUM PROFILE: 10 NM X-OVERLAY

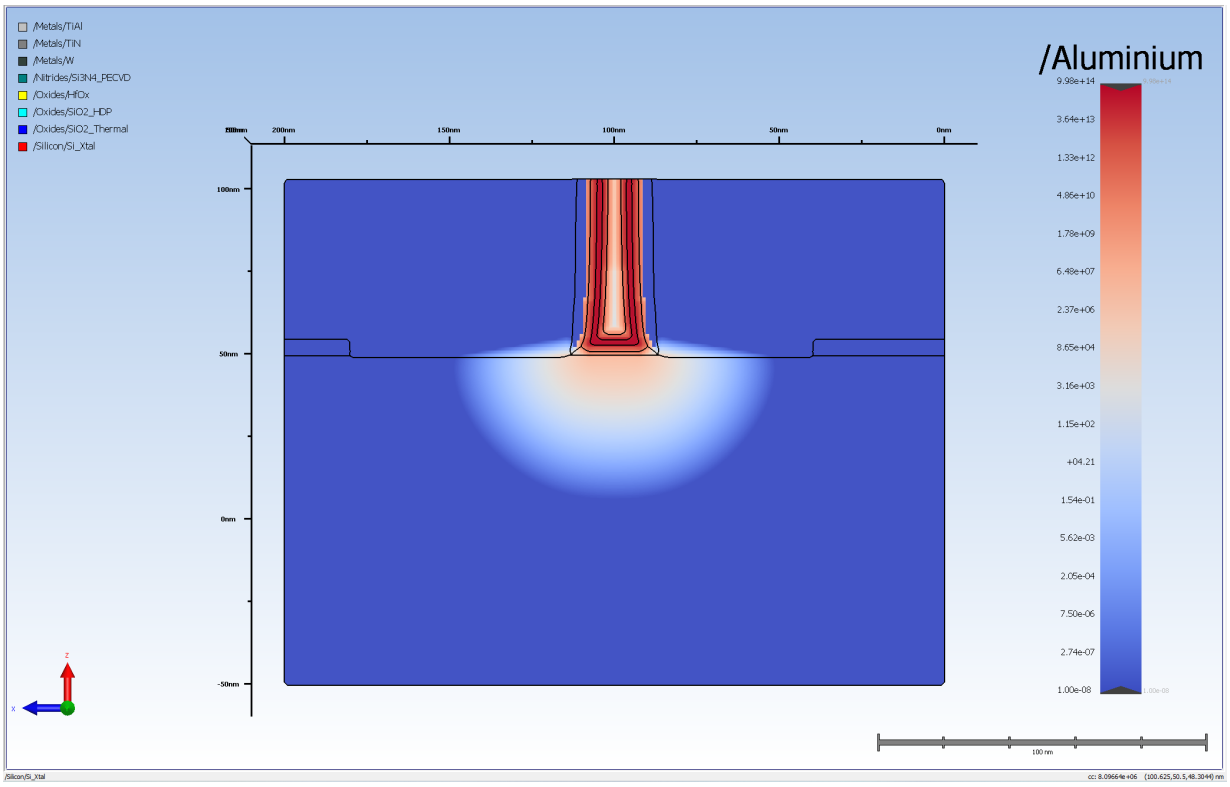


Figure 7: Aluminium Concentration Field After Source_Drain2 Mask & 10 nm X-Overlay